Python Script Used for Bit-Flip Fault

```
from pygdbmi.gdbcontroller import GdbController
from docx import Document
from datetime import datetime
import time

# Critical registers to test
registers_to_test = ['pc', 'x1', 'x2', 'x8', 'x9']
bit_to_flip = 63  # MSB for 64-bit registers
iterations = 2
delay_seconds = 1

# Setup GDB MI controller
gdbmi = GdbController(command=["gdb-multiarch", "--nx", "--interpreter=mi3"])
gdbmi.write('-target-select remote localhost:1234')
time.sleep(1)

# Simulate background load in guest
print("[*] Starting background load loop in guest OS...")
gdbmi.write(
    'call (int)system("while true; do echo alive >> /tmp/heartbeat.log; sleep 1; done &")'
)
time.sleep(1)

# Prepare doc report
doc = Document()
doc.add_heading('Bit Flip Fault Injection Report (MSB)', 0)
doc.add_paragraph(f'Test started: {datetime.now()}')
doc.add_paragraph(f'Registers tested: {", ".join(registers_to_test)}')
doc.add_paragraph(f'Bit flipped: {bit_to_flip} (MSB)')
doc.add_paragraph(f'Iterations: {iterations}')
doc.add_paragraph('')

for i in range(iterations):
    doc.add_heading(f'Iteration {i+1}', level=1)

    # Interrupt GDB target to manipulate registers
    gdbmi.write('interrupt')
    time.sleep(0.5)

    for reg in registers_to_test:
        resp = gdbmi.write(f'-data-evaluate-expression ${reg}')
        current_val = None
        for r in resp:
            if r['type'] == 'result' and 'payload' in r:
                val = r['payload'].get('value')
                if val:
                    try:
                        current_val = int(val, 16)
                    except ValueError:
                        current_val = None
                break

        if current_val is None:
```

```
            doc.add_paragraph(f'[-] Could not read register {reg}')
            continue

        flipped_val = current_val ^ (1 << bit_to_flip)
        gdbmi.write(f'set ${reg} = {flipped_val}')
        doc.add_paragraph(
            f'{reg}: initial=0x{current_val:x}, flipped MSB (bit 63) → 0x{flipped_val:x}'
        )

    doc.add_paragraph('')
    time.sleep(delay_seconds)

report_path = '/mnt/f/loggins/bit_flip_msb_critical_regs.docx'
doc.save(report_path)

print(f'[+] Completed {iterations} iterations. Report saved at {report_path}')
gdbmi.exit()
```

Bit Flip Fault Injection Report (MSB)

Test started: 2025-05-23 17:46:03.449302

Registers tested: pc, x1, x2, x8, x9

Bit flipped: 63 (MSB)

Iterations: 2

Iteration 1

pc: initial=0x1000, flipped MSB (bit 63) → 0x8000000000001000

x1: initial=0x0, flipped MSB (bit 63) → 0x8000000000000000

x2: initial=0x0, flipped MSB (bit 63) → 0x8000000000000000

x8: initial=0x0, flipped MSB (bit 63) → 0x8000000000000000

x9: initial=0x0, flipped MSB (bit 63) → 0x8000000000000000

Iteration 2

pc: initial=0x8000000000001000, flipped MSB (bit 63) → 0x1000

x1: initial=0x8000000000000000, flipped MSB (bit 63) → 0x0

x2: initial=0x8000000000000000, flipped MSB (bit 63) → 0x0

x8: initial=0x8000000000000000, flipped MSB (bit 63) → 0x0

x9: initial=0x-9223372036854775808, flipped MSB (bit 63) → 0x-922b372036854775808

Console Log of the Bit-Flip Fault Injection Run

```
OpenSBI v1.5.1
   ____                    _____ ____ _____
  / __ \                  / ____|  _ \_   _|
 | |  | |_ __   ___ _ __ | (___ | |_) || |
 | |  | | '_ \ / _ \ '_ \ \___ \|  _ < | |
 | |__| | |_) |  __/ | | |____) | |_) || |_
  \____/| .__/ \___|_| |_|_____/|____/_____|
        | |
        |_|

Platform Name             : riscv-virtio,qemu
Platform Features         : medeleg
Platform HART Count       : 1
Platform IPI Device       : aclint-mswi
Platform Timer Device     : aclint-mtimer @ 10000000Hz
Platform Console Device   : uart8250
Platform HSM Device       : ---
Platform PMU Device       : ---
Platform Reboot Device    : syscon-reboot
Platform Shutdown Device  : syscon-poweroff
Platform Suspend Device   : ---
Platform CPPC Device      : ---
Firmware Base             : 0x80000000
Firmware Size             : 327 KB
Firmware RW Offset        : 0x40000
Firmware RW Size          : 71 KB
Firmware Heap Offset      : 0x49000
Firmware Heap Size        : 35 KB (total), 2 KB (reserved), 11 KB (used), 21 KB
(free)
Firmware Scratch Size     : 4096 B (total), 416 B (used), 3680 B (free)
Runtime SBI Version       : 2.0

Domain0 Name              : root
Domain0 Boot HART         : 0
Domain0 HARTs             : 0*
Domain0 Region00          : 0x0000000000100000-0x0000000000100fff M: (I,R,W) S/U:
(R,W)
Domain0 Region01          : 0x0000000010000000-0x0000000010000fff M: (I,R,W) S/U:
(R,W)
Domain0 Region02          : 0x0000000002000000-0x000000000200ffff M: (I,R,W) S/U:
()
Domain0 Region03          : 0x0000000080040000-0x000000008005ffff M: (R,W) S/U: ()
Domain0 Region04          : 0x0000000080000000-0x000000008003ffff M: (R,X) S/U: ()
Domain0 Region05          : 0x000000000c400000-0x000000000c5fffff M: (I,R,W) S/U:
(R,W)
Domain0 Region06          : 0x000000000c000000-0x000000000c3fffff M: (I,R,W) S/U:
(R,W)
Domain0 Region07          : 0x0000000000000000-0xffffffffffffffff M: () S/U:
(R,W,X)
Domain0 Next Address      : 0x0000000080200000
Domain0 Next Arg1         : 0x0000000087e00000
Domain0 Next Mode         : S-mode
Domain0 SysReset          : yes
Domain0 SysSuspend        : yes

Boot HART ID              : 0
Boot HART Domain          : root
Boot HART Priv Version    : v1.12
Boot HART Base ISA        : rv64imafdch
Boot HART ISA Extensions  : sstc,zicntr,zihpm,zicboz,zicbom,sdtrig,svadu
Boot HART PMP Count       : 16
Boot HART PMP Granularity : 2 bits
Boot HART PMP Address Bits: 54
```

```
Boot HART MHPM Info       : 16 (0x0007fff8)
Boot HART Debug Triggers  : 2 triggers
Boot HART MIDELEG         : 0x0000000000001666
Boot HART MEDELEG         : 0x0000000000f0b509
[    0.000000] Linux version 6.14.5 (kazumin@PR3D4T0R) (riscv64-linux-gnu-gcc
(Ubuntu 13.3.0-6ubuntu2~24.04) 13.3.0, GNU ld (GNU Binutils for Ubuntu) 2.42) #1
SMP Fri May  2 19:37:40 CEST 2025
[    0.000000] random: crng init done
[    0.000000] Machine model: riscv-virtio,qemu
[    0.000000] SBI specification v2.0 detected
[    0.000000] SBI implementation ID=0x1 Version=0x10005
[    0.000000] SBI TIME extension detected
[    0.000000] SBI IPI extension detected
[    0.000000] SBI RFENCE extension detected
[    0.000000] SBI SRST extension detected
[    0.000000] SBI DBCN extension detected
[    0.000000] efi: UEFI not found.
[    0.000000] OF: reserved mem: 0x0000000080000000..0x000000008003ffff (256 KiB)
nomap non-reusable mmode_resv1@80000000
[    0.000000] OF: reserved mem: 0x0000000080040000..0x000000008005ffff (128 KiB)
nomap non-reusable mmode_resv0@80040000
[    0.000000] Zone ranges:
[    0.000000]   DMA32    [mem 0x0000000080000000-0x0000000087ffffff]
[    0.000000]   Normal   empty
[    0.000000] Movable zone start for each node
[    0.000000] Early memory node ranges
[    0.000000]   node   0: [mem 0x0000000080000000-0x000000008005ffff]
[    0.000000]   node   0: [mem 0x0000000080060000-0x0000000087ffffff]
[    0.000000] Initmem setup node 0 [mem 0x0000000080000000-0x0000000087ffffff]
[    0.000000] SBI HSM extension detected
[    0.000000] riscv: base ISA extensions acdfhim
[    0.000000] riscv: ELF capabilities acdfim
[    0.000000] Queued spinlock using Ziccrse: enabled
[    0.000000] percpu: Embedded 22 pages/cpu s49448 r8192 d32472 u90112
[    0.000000] Kernel command line: root=/dev/vda1 console=ttyS0
[    0.000000] printk: log buffer data + meta data: 131072 + 458752 = 589824 bytes
[    0.000000] Dentry cache hash table entries: 16384 (order: 5, 131072 bytes,
linear)
[    0.000000] Inode-cache hash table entries: 8192 (order: 4, 65536 bytes, linear)
[    0.000000] Built 1 zonelists, mobility grouping on.  Total pages: 32768
[    0.000000] mem auto-init: stack:all(zero), heap alloc:off, heap free:off
[    0.000000] software IO TLB: SWIOTLB bounce buffer size adjusted to 0MB
[    0.000000] software IO TLB: area num 1.
[    0.000000] software IO TLB: mapped [mem 0x0000000087f65000-0x0000000087fa5000]
(0MB)
[    0.000000] Virtual kernel memory layout:
[    0.000000]       fixmap : 0xff1bfffffea00000 - 0xff1bffffff000000   (6144 kB)
[    0.000000]       pci io : 0xff1bffffff000000 - 0xff1c000000000000   (  16 MB)
[    0.000000]      vmemmap : 0xff1c000000000000 - 0xff20000000000000   (1024 TB)
[    0.000000]      vmalloc : 0xff20000000000000 - 0xff60000000000000   (16384 TB)
[    0.000000]      modules : 0xffffffff0158f000 - 0xffffffff80000000   (2026 MB)
[    0.000000]       lowmem : 0xff60000000000000 - 0xff60000008000000   ( 128 MB)
[    0.000000]       kernel : 0xffffffff80000000 - 0xffffffffffffffff   (2047 MB)
[    0.000000] SLUB: HWalign=64, Order=0-3, MinObjects=0, CPUs=1, Nodes=1
[    0.000000] rcu: Hierarchical RCU implementation.
[    0.000000] rcu:    RCU restricting CPUs from NR_CPUS=64 to nr_cpu_ids=1.
[    0.000000] rcu:    RCU debug extended QS entry/exit.
[    0.000000]   Tracing variant of Tasks RCU enabled.
[    0.000000] rcu: RCU calculated value of scheduler-enlistment delay is 25
jiffies.
[    0.000000] rcu: Adjusting geometry for rcu_fanout_leaf=16, nr_cpu_ids=1
[    0.000000] RCU Tasks Trace: Setting shift to 0 and lim to 1
rcu_task_cb_adjust=1 rcu_task_cpu_ids=1.
[    0.000000] NR_IRQS: 64, nr_irqs: 64, preallocated irqs: 0
[    0.000000] riscv-intc: 64 local interrupts mapped
[    0.000000] riscv: providing IPIs using SBI IPI extension
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[    0.000000] rcu: srcu_init: Setting srcu_struct sizes based on contention.
[    0.000000] clocksource: riscv_clocksource: mask: 0xffffffffffffffff max_cycles:
0x24e6a1710, max_idle_ns: 440795202120 ns
[    0.000070] sched_clock: 64 bits at 10MHz, resolution 100ns, wraps every
4398046511100ns
[    0.000159] riscv-timer: Timer interrupt in S-mode is available via sstc
extension
[    0.005017] Console: colour dummy device 80x25
[    0.008408] Calibrating delay loop (skipped), value calculated using timer
frequency.. 20.00 BogoMIPS (lpj=40000)
[    0.008511] pid_max: default: 32768 minimum: 301
[    0.009735] LSM: initializing lsm=capability
[    0.011960] Mount-cache hash table entries: 512 (order: 0, 4096 bytes, linear)
[    0.011990] Mountpoint-cache hash table entries: 512 (order: 0, 4096 bytes,
linear)
[    0.041821] riscv: ELF compat mode supported
[    0.042002] ASID allocator using 16 bits (65536 entries)
[    0.043129] rcu: Hierarchical SRCU implementation.
[    0.043157] rcu:     Max phase no-delay instances is 1000.
[    0.051112] EFI services will not be available.
[    0.054088] smp: Bringing up secondary CPUs ...
[    0.054554] smp: Brought up 1 node, 1 CPU
[    0.057412] Memory: 87492K/131072K available (10189K kernel code, 4941K rwdata,
4096K rodata, 2307K init, 484K bss, 41972K reserved, 0K cma-reserved)
[    0.063439] devtmpfs: initialized
[    0.073211] clocksource: jiffies: mask: 0xffffffff max_cycles: 0xffffffff,
max_idle_ns: 7645041785100000 ns
[    0.073347] futex hash table entries: 256 (order: 2, 16384 bytes, linear)
[    0.075841] pinctrl core: initialized pinctrl subsystem
[    0.082218] DMI not present or invalid.
[    0.083709] NET: Registered PF_NETLINK/PF_ROUTE protocol family
[    0.086648] DMA: preallocated 128 KiB GFP_KERNEL pool for atomic allocations
[    0.086879] DMA: preallocated 128 KiB GFP_KERNEL|GFP_DMA32 pool for atomic
allocations
[    0.087068] audit: initializing netlink subsys (disabled)
[    0.090088] thermal_sys: Registered thermal governor 'step_wise'
[    0.090827] audit: type=2000 audit(0.080:1): state=initialized audit_enabled=0
res=1
[    0.091317] cpuidle: using governor menu
[    0.114563] cpu0: Ratio of byte access time to unaligned word access is 6.93,
unaligned accesses are fast
[    0.132805] HugeTLB: registered 2.00 MiB page size, pre-allocated 0 pages
[    0.132829] HugeTLB: 28 KiB vmemmap can be freed for a 2.00 MiB page
[    0.136809] ACPI: Interpreter disabled.
[    0.137184] iommu: Default domain type: Translated
[    0.137209] iommu: DMA domain TLB invalidation policy: strict mode
[    0.139000] SCSI subsystem initialized
[    0.140417] usbcore: registered new interface driver usbfs
[    0.140625] usbcore: registered new interface driver hub
[    0.140736] usbcore: registered new device driver usb
[    0.142537] Advanced Linux Sound Architecture Driver Initialized.
[    0.152926] vgaarb: loaded
[    0.154906] clocksource: Switched to clocksource riscv_clocksource
[    0.158087] pnp: PnP ACPI: disabled
[    0.175923] NET: Registered PF_INET protocol family
[    0.176685] IP idents hash table entries: 2048 (order: 2, 16384 bytes, linear)
[    0.181159] tcp_listen_portaddr_hash hash table entries: 128 (order: 0, 4096
bytes, linear)
[    0.181237] Table-perturb hash table entries: 65536 (order: 6, 262144 bytes,
linear)
[    0.181308] TCP established hash table entries: 1024 (order: 1, 8192 bytes,
linear)
[    0.181419] TCP bind hash table entries: 1024 (order: 4, 65536 bytes, linear)
[    0.181584] TCP: Hash tables configured (established 1024 bind 1024)
[    0.182268] UDP hash table entries: 256 (order: 3, 40960 bytes, linear)
[    0.182616] UDP-Lite hash table entries: 256 (order: 3, 40960 bytes, linear)
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[    0.183591] NET: Registered PF_UNIX/PF_LOCAL protocol family
[    0.185768] RPC: Registered named UNIX socket transport module.
[    0.185804] RPC: Registered udp transport module.
[    0.185812] RPC: Registered tcp transport module.
[    0.185819] RPC: Registered tcp-with-tls transport module.
[    0.185826] RPC: Registered tcp NFSv4.1 backchannel transport module.
[    0.185939] PCI: CLS 0 bytes, default 64
[    0.192197] workingset: timestamp_bits=46 max_order=15 bucket_order=0
[    0.196228] NFS: Registering the id_resolver key type
[    0.196999] Key type id_resolver registered
[    0.197027] Key type id_legacy registered
[    0.197279] nfs4filelayout_init: NFSv4 File Layout Driver Registering...
[    0.197355] nfs4flexfilelayout_init: NFSv4 Flexfile Layout Driver Registering...
[    0.197811] 9p: Installing v9fs 9p2000 file system support
[    0.199282] NET: Registered PF_ALG protocol family
[    0.199523] Block layer SCSI generic (bsg) driver version 0.4 loaded (major 245)
[    0.199928] io scheduler mq-deadline registered
[    0.200000] io scheduler kyber registered
[    0.200197] io scheduler bfq registered
[    0.202978] riscv-plic: plic@c000000: mapped 95 interrupts with 1 handlers for 2
contexts.
[    0.205538] pci-host-generic 30000000.pci: host bridge /soc/pci@30000000 ranges:
[    0.206116] pci-host-generic 30000000.pci:       IO 0x0003000000..0x000300ffff -
> 0x0000000000
[    0.206577] pci-host-generic 30000000.pci:      MEM 0x0040000000..0x007fffffff -
> 0x0040000000
[    0.206634] pci-host-generic 30000000.pci:      MEM 0x0400000000..0x07ffffffff -
> 0x0400000000
[    0.207032] pci-host-generic 30000000.pci: Memory resource size exceeds max for
32 bits
[    0.207372] pci-host-generic 30000000.pci: ECAM at [mem 0x30000000-0x3fffffff]
for [bus 00-ff]
[    0.208558] pci-host-generic 30000000.pci: PCI host bridge to bus 0000:00
[    0.208812] pci_bus 0000:00: root bus resource [bus 00-ff]
[    0.208880] pci_bus 0000:00: root bus resource [io  0x0000-0xffff]
[    0.208913] pci_bus 0000:00: root bus resource [mem 0x40000000-0x7fffffff]
[    0.208932] pci_bus 0000:00: root bus resource [mem 0x400000000-0x7ffffffff]
[    0.210045] pci 0000:00:00.0: [1b36:0008] type 00 class 0x060000 conventional
PCI endpoint
[    0.213143] pci 0000:00:01.0: [1af4:1001] type 00 class 0x010000 conventional
PCI endpoint
[    0.213334] pci 0000:00:01.0: BAR 0 [io  0x0000-0x007f]
[    0.213368] pci 0000:00:01.0: BAR 1 [mem 0x00000000-0x00000fff]
[    0.213404] pci 0000:00:01.0: BAR 4 [mem 0x00000000-0x00003fff 64bit pref]
[    0.215140] pci 0000:00:01.0: BAR 4 [mem 0x400000000-0x400003fff 64bit pref]:
assigned
[    0.215333] pci 0000:00:01.0: BAR 1 [mem 0x40000000-0x40000fff]: assigned
[    0.215362] pci 0000:00:01.0: BAR 0 [io  0x0080-0x00ff]: assigned
[    0.215591] pci_bus 0000:00: resource 4 [io  0x0000-0xffff]
[    0.215618] pci_bus 0000:00: resource 5 [mem 0x40000000-0x7fffffff]
[    0.215627] pci_bus 0000:00: resource 6 [mem 0x400000000-0x7ffffffff]
[    0.220040] SBI CPPC extension NOT detected!!
[    0.227146] virtio-pci 0000:00:01.0: enabling device (0000 -> 0003)
[    0.303569] Serial: 8250/16550 driver, 4 ports, IRQ sharing disabled
[    0.313052] printk: legacy console [ttyS0] disabled
[    0.315637] 10000000.serial: ttyS0 at MMIO 0x10000000 (irq = 13, base_baud =
230400) is a 16550A
[    0.316955] printk: legacy console [ttyS0] enabled
[    1.606470] loop: module loaded
[    1.611839] virtio_blk virtio0: 1/0/0 default/read/poll queues
[    1.623156] virtio_blk virtio0: [vda] 9437184 512-byte logical blocks (4.83
GB/4.50 GiB)
[    1.653528]  vda: vda1 vda12 vda13 vda14 vda15 vda16
[    1.669693] e1000e: Intel(R) PRO/1000 Network Driver
[    1.676705] e1000e: Copyright(c) 1999 - 2015 Intel Corporation.
[    1.686935] usbcore: registered new interface driver uas
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[    1.695161] usbcore: registered new interface driver usb-storage
[    1.704412] mousedev: PS/2 mouse device common for all mice
[    1.714609] goldfish_rtc 101000.rtc: registered as rtc0
[    1.722157] goldfish_rtc 101000.rtc: setting system clock to 2025-05-24T07:46:24
UTC (1748072784)
[    1.738453] sdhci: Secure Digital Host Controller Interface driver
[    1.746759] sdhci: Copyright(c) Pierre Ossman
[    1.752808] Synopsys Designware Multimedia Card Interface Driver
[    1.761396] sdhci-pltfm: SDHCI platform and OF driver helper
[    1.769898] usbcore: registered new interface driver usbhid
[    1.777656] usbhid: USB HID core driver
[    1.783110] riscv-pmu-sbi: SBI PMU extension is available
[    1.790980] riscv-pmu-sbi: 16 firmware and 18 hardware counters
[    1.798870] riscv-pmu-sbi: Perf sampling/filtering is not supported as sscof
extension is not available
[    1.816392] NET: Registered PF_INET6 protocol family
[    1.828685] Segment Routing with IPv6
[    1.833690] In-situ OAM (IOAM) with IPv6
[    1.839355] sit: IPv6, IPv4 and MPLS over IPv4 tunneling driver
[    1.849522] NET: Registered PF_PACKET protocol family
[    1.857060] 9pnet: Installing 9P2000 support
[    1.862765] Key type dns_resolver registered
[    1.900189] debug_vm_pgtable: [debug_vm_pgtable         ]: Validating
architecture page table helpers
[    1.920899] clk: Disabling unused clocks
[    1.926236] PM: genpd: Disabling unused power domains
[    1.933197] ALSA device list:
[    1.937219]   No soundcards found.
[    1.967880] EXT4-fs (vda1): INFO: recovery required on readonly filesystem
[    1.977257] EXT4-fs (vda1): write access will be enabled during recovery
[    2.006688] EXT4-fs (vda1): recovery complete
[    2.015296] EXT4-fs (vda1): mounted filesystem 626ef4b6-84d2-4b70-bd20-
e9f892339465 ro with ordered data mode. Quota mode: disabled.
[    2.031304] VFS: Mounted root (ext4 filesystem) readonly on device 254:1.
[    2.043596] devtmpfs: mounted
[    2.086962] Freeing unused kernel image (initmem) memory: 2304K
[    2.095959] Run /sbin/init as init process
[    2.359682] systemd[1]: systemd 255.4-1ubuntu8.5 running in system mode (+PAM
+AUDIT +SELINUX +APPARMOR +IMA +SMACK +SECCOMP +GCRYPT -GNUTLS +OPENSSL +ACL +BLKID
+CURL +ELFUTILS +FIDO2 +IDN2 -IDN +IPTC +KMOD +LIBCRYPTSETUP +LIBFDISK +PCRE2 -
PWQUALITY +P11KIT +QRENCODE +TPM2 +BZIP2 +LZ4 +XZ +ZLIB +ZSTD -BPF_FRAMEWORK -
XKBCOMMON +UTMP +SYSVINIT default-hierarchy=unified)
[    2.405123] systemd[1]: Detected virtualization qemu.
[    2.412211] systemd[1]: Detected architecture riscv64.

Welcome to [1mUbuntu 24.04.2 LTS[0m!



[    2.433388] systemd[1]: Hostname set to <ubuntu>.
[    4.530306] systemd[1]: Queued start job for default target graphical.target.
[    4.593010] systemd[1]: Created slice system-modprobe.slice - Slice
/system/modprobe.
[[0;32m  OK  [0m] Created slice [0;1;39msystem-modprobe.slice[0m - Slice
/system/modprobe.

[    4.621933] systemd[1]: Created slice system-serial\x2dgetty.slice - Slice
/system/serial-getty.
[[0;32m  OK  [0m] Created slice [0;1;39msystem-serial\x2dgetâ€¦slice[0m - Slice
/system/serial-getty.

[    4.650786] systemd[1]: Created slice system-systemd\x2dfsck.slice - Slice
/system/systemd-fsck.
[[0;32m  OK  [0m] Created slice [0;1;39msystem-systemd\x2dfsâ€¦slice[0m - Slice
/system/systemd-fsck.

[    4.678207] systemd[1]: Created slice user.slice - User and Session Slice.
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[[0;32m OK [0m] Created slice [0;1;39muser.slice[0m - User and Session Slice.

[4.700288] systemd[1]: Started systemd-ask-password-wall.path - Forward Password Requests to Wall Directory Watch.
[[0;32m OK [0m] Started [0;1;39msystemd-ask-password-wall.â€¦[0md Requests to Wall Directory Watch.

[4.728093] systemd[1]: proc-sys-fs-binfmt_misc.automount - Arbitrary Executable File Formats File System Automount Point was skipped because of an unmet condition check (ConditionPathExists=/proc/sys/fs/binfmt_misc).
[4.754990] systemd[1]: Expecting device dev-disk-by\x2dlabel-UEFI.device - /dev/disk/by-label/UEFI...
         Expecting device [0;1;39mdev-disk-by\x2dlaâ€¦device[0m - /dev/disk/by-label/UEFI...

[4.779530] systemd[1]: Expecting device dev-ttyS0.device - /dev/ttyS0...
         Expecting device [0;1;39mdev-ttyS0.device[0m - /dev/ttyS0...

[4.797541] systemd[1]: Reached target integritysetup.target - Local Integrity Protected Volumes.
[[0;32m OK [0m] Reached target [0;1;39mintegritysetup.targâ€¦[0m Local Integrity Protected Volumes.

[4.822629] systemd[1]: Reached target slices.target - Slice Units.
[[0;32m OK [0m] Reached target [0;1;39mslices.target[0m - Slice Units.

[4.840598] systemd[1]: Reached target snapd.mounts-pre.target - Mounting snaps.
[[0;32m OK [0m] Reached target [0;1;39msnapd.mounts-pre.target[0m - Mounting snaps.

[4.861589] systemd[1]: Reached target snapd.mounts.target - Mounted snaps.
[[0;32m OK [0m] Reached target [0;1;39msnapd.mounts.target[0m - Mounted snaps.

[4.881495] systemd[1]: Reached target swap.target - Swaps.
[[0;32m OK [0m] Reached target [0;1;39mswap.target[0m - Swaps.

[4.898121] systemd[1]: Reached target veritysetup.target - Local Verity Protected Volumes.
[[0;32m OK [0m] Reached target [0;1;39mveritysetup.target[0m - Local Verity Protected Volumes.

[4.924554] systemd[1]: Listening on dm-event.socket - Device-mapper event daemon FIFOs.
[[0;32m OK [0m] Listening on [0;1;39mdm-event.socket[0m - Device-mapper event daemon FIFOs.

[4.948231] systemd[1]: Listening on lvm2-lvmpolld.socket - LVM2 poll daemon socket.
[[0;32m OK [0m] Listening on [0;1;39mlvm2-lvmpolld.socket[0m - LVM2 poll daemon socket.

[4.971654] systemd[1]: Listening on multipathd.socket - multipathd control socket.
[[0;32m OK [0m] Listening on [0;1;39mmultipathd.socket[0m - multipathd control socket.

[4.994160] systemd[1]: Listening on syslog.socket - Syslog Socket.
[[0;32m OK [0m] Listening on [0;1;39msyslog.socket[0m - Syslog Socket.

[5.013008] systemd[1]: Listening on systemd-fsckd.socket - fsck to fsckd communication Socket.
[[0;32m OK [0m] Listening on [0;1;39msystemd-fsckd.socket[0m â€¦fsck to fsckd communication Socket.

[5.037966] systemd[1]: Listening on systemd-initctl.socket - initctl Compatibility Named Pipe.

```
[[0;32m  OK  [0m] Listening on [0;1;39msystemd-initctl.socke…[0m- initctl
Compatibility Named Pipe.

[    5.063907] systemd[1]: Listening on systemd-journald-dev-log.socket - Journal
Socket (/dev/log).
[[0;32m  OK  [0m] Listening on [0;1;39msystemd-journald-dev-…socket[0m - Journal
Socket (/dev/log).

[    5.090458] systemd[1]: Listening on systemd-journald.socket - Journal Socket.
[[0;32m  OK  [0m] Listening on [0;1;39msystemd-journald.socket[0m - Journal Socket.

[    5.113760] systemd[1]: Listening on systemd-networkd.socket - Network Service
Netlink Socket.
[[0;32m  OK  [0m] Listening on [0;1;39msystemd-networkd.socket[0m - Network Service
Netlink Socket.

[    5.139407] systemd[1]: systemd-pcrextend.socket - TPM2 PCR Extension (Varlink)
was skipped because of an unmet condition check (ConditionSecurity=measured-uki).
[    5.160942] systemd[1]: Listening on systemd-udevd-control.socket - udev Control
Socket.
[[0;32m  OK  [0m] Listening on [0;1;39msystemd-udevd-control.socket[0m - udev
Control Socket.

[    5.186213] systemd[1]: Listening on systemd-udevd-kernel.socket - udev Kernel
Socket.
[[0;32m  OK  [0m] Listening on [0;1;39msystemd-udevd-kernel.socket[0m - udev Kernel
Socket.

[    5.235895] systemd[1]: Mounting dev-hugepages.mount - Huge Pages File System...
         Mounting [0;1;39mdev-hugepages.mount[0m - Huge Pages File System...

[    5.306973] systemd[1]: Mounting dev-mqueue.mount - POSIX Message Queue File
System...
         Mounting [0;1;39mdev-mqueue.mount[0m - POSIX Message Queue File System...

[    5.399125] systemd[1]: Mounting sys-kernel-debug.mount - Kernel Debug File
System...
         Mounting [0;1;39msys-kernel-debug.mount[0m - Kernel Debug File System...

[    5.447591] systemd[1]: sys-kernel-tracing.mount - Kernel Trace File System was
skipped because of an unmet condition check
(ConditionPathExists=/sys/kernel/tracing).
[    5.529029] systemd[1]: Starting systemd-journald.service - Journal Service...
         Starting [0;1;39msystemd-journald.service[0m - Journal Service...

[    5.603469] systemd[1]: Starting keyboard-setup.service - Set the console
keyboard layout...
         Starting [0;1;39mkeyboard-setup.service[0m - Set the console keyboard
layout...

[    5.664165] systemd[1]: kmod-static-nodes.service - Create List of Static Device
Nodes was skipped because of an unmet condition check
(ConditionFileNotEmpty=/lib/modules/6.14.5/modules.devname).
[    5.775737] systemd[1]: Starting lvm2-monitor.service - Monitoring of LVM2
mirrors, snapshots etc. using dmeventd or progress polling...
         Starting [0;1;39mlvm2-monitor.service[0m - Mo…ing dmeventd or progress
polling...

[    5.936759] systemd-journald[88]: Collecting audit messages is disabled.
[    5.959700] systemd[1]: Starting modprobe@configfs.service - Load Kernel Module
configfs...
         Starting [0;1;39mmodprobe@configfs.service[0m - Load Kernel Module
configfs...

[    6.131584] systemd[1]: Starting modprobe@dm_mod.service - Load Kernel Module
dm_mod...
```

```
         Starting [0;1;39mmodprobe@dm_mod.service[0m - Load Kernel Module dm_mod...

[    6.331501] systemd[1]: Starting modprobe@drm.service - Load Kernel Module
drm...
         Starting [0;1;39mmodprobe@drm.service[0m - Load Kernel Module drm...

[    6.503529] systemd[1]: Starting modprobe@efi_pstore.service - Load Kernel
Module efi_pstore...
         Starting [0;1;39mmodprobe@efi_pstore.serviâ€¦[0m - Load Kernel Module
efi_pstore...

[    6.651375] systemd[1]: Starting modprobe@fuse.service - Load Kernel Module
fuse...
         Starting [0;1;39mmodprobe@fuse.service[0m - Load Kernel Module fuse...

[    6.767596] systemd[1]: Starting modprobe@loop.service - Load Kernel Module
loop...
         Starting [0;1;39mmodprobe@loop.service[0m - Load Kernel Module loop...

[    6.847130] systemd[1]: netplan-ovs-cleanup.service - OpenVSwitch configuration
for cleanup was skipped because of an unmet condition check
(ConditionFileIsExecutable=/usr/bin/ovs-vsctl).
[    6.939603] systemd[1]: Starting systemd-fsck-root.service - File System Check
on Root Device...
         Starting [0;1;39msystemd-fsck-root.service[0mâ€¦File System Check on Root
Device...

[    7.059453] systemd[1]: Starting systemd-modules-load.service - Load Kernel
Modules...
         Starting [0;1;39msystemd-modules-load.service[0m - Load Kernel Modules...

[    7.140872] systemd[1]: systemd-pcrmachine.service - TPM2 PCR Machine ID
Measurement was skipped because of an unmet condition check
(ConditionSecurity=measured-uki).
[    7.275428] systemd[1]: Starting systemd-tmpfiles-setup-dev-early.service -
Create Static Device Nodes in /dev gracefully...
         Starting [0;1;39msystemd-tmpfiles-setup-deâ€¦[0m Device Nodes in /dev
gracefully...

[    7.390300] systemd[1]: systemd-tpm2-setup-early.service - TPM2 SRK Setup
(Early) was skipped because of an unmet condition check
(ConditionSecurity=measured-uki).
[    7.487607] systemd[1]: Starting systemd-udev-trigger.service - Coldplug All
udev Devices...
         Starting [0;1;39msystemd-udev-trigger.service[0m - Coldplug All udev
Devices...

[    7.636162] systemd[1]: Started systemd-journald.service - Journal Service.
[[0;32m  OK  [0m] Started [0;1;39msystemd-journald.service[0m - Journal Service.

[[0;32m  OK  [0m] Mounted [0;1;39mdev-hugepages.mount[0m - Huge Pages File System.

[[0;32m  OK  [0m] Mounted [0;1;39mdev-mqueue.mount[0m - POSIX Message Queue
```

Python Script Used for Stuck-at Fault

```
from pygdbmi.gdbcontroller import GdbController
from docx import Document
from datetime import datetime
import time

# === Configurable params ===
# Critical registers for automotive safety
registers_to_test = ['pc', 'x1', 'x2', 'x8', 'x9']
iterations_per_mode = 2
delay_seconds = 1
msb_bit = 63   # Most significant bit for 64-bit registers

# === Setup GDB MI controller ===
gdbmi = GdbController(command=["gdb-multiarch", "--nx", "--interpreter=mi3"])
gdbmi.write('-target-select remote localhost:1234')
time.sleep(1)

# === Kick off a background load in the guest ===
print("[*] Starting background load loop in guest OS...")
gdbmi.write(
    'call (int)system("while true; do echo alive >> /tmp/heartbeat.log; sleep 1; done &")'
)
time.sleep(1)

# === Prepare report ===
doc = Document()
doc.add_heading('Stuck-At Fault Injection Report', 0)
doc.add_paragraph(f'Test started: {datetime.now()}')
doc.add_paragraph(f'Registers tested: {", ".join(registers_to_test)}')
doc.add_paragraph(f'Iterations per mode: {iterations_per_mode}')
doc.add_paragraph('Fault modes: stuck-at-1 then stuck-at-0 (MSB)')
doc.add_paragraph('')

def safe_interrupt():
    """Try interrupting the CPU; retry up to 3 times."""
    for _ in range(3):
        try:
            return gdbmi.write('interrupt')
        except Exception:
            time.sleep(1)
    raise Exception("GDB did not respond to interrupt after retries.")

msb_mask = 1 << msb_bit

# === Stuck-at-1 on MSB ===
for i in range(iterations_per_mode):
    doc.add_heading(f'Iteration stuck-at-1 (MSB): {i+1}', level=1)
    safe_interrupt()
    time.sleep(0.5)

    for reg in registers_to_test:
        # Read current value
        resp = gdbmi.write(f'-data-evaluate-expression ${reg}')
```

```
        current = None
        for r in resp:
            if r['type'] == 'result' and 'payload' in r:
                val = r['payload'].get('value')
                if val:
                    try:
                        current = int(val, 16)
                    except ValueError:
                        current = None
                break

        if current is None:
            doc.add_paragraph(f'[-] Could not read register {reg}')
            continue

        # Force MSB to 1
        stuck_val = current | msb_mask
        gdbmi.write(f'set ${reg} = {stuck_val}')

        doc.add_paragraph(
            f'{reg}: initial=0x{current:x}, stuck-at-1 MSB → 0x{stuck_val:x}'
        )

    doc.add_paragraph('')
    time.sleep(delay_seconds)

# === Stuck-at-0 on MSB ===
for i in range(iterations_per_mode):
    doc.add_heading(f'Iteration stuck-at-0 (MSB): {i+1}', level=1)
    safe_interrupt()
    time.sleep(0.5)

    for reg in registers_to_test:
        resp = gdbmi.write(f'-data-evaluate-expression ${reg}')
        current = None
        for r in resp:
            if r['type'] == 'result' and 'payload' in r:
                val = r['payload'].get('value')
                if val:
                    try:
                        current = int(val, 16)
                    except ValueError:
                        current = None
                break

        if current is None:
            doc.add_paragraph(f'[-] Could not read register {reg}')
            continue

        # Force MSB to 0
        stuck_val = current & ~msb_mask
        gdbmi.write(f'set ${reg} = {stuck_val}')

        doc.add_paragraph(
```

```
            f'{reg}: initial=0x{current:x}, stuck-at-0 MSB → 0x{stuck_val:x}'
        )

    doc.add_paragraph('')
    time.sleep(delay_seconds)

# === Save report ===
report_path = '/mnt/f/loggins_stuck_at/stuck_at_fault_report.docx'
doc.save(report_path)
print(f'[+] Completed stuck-at MSB faults. Report saved at {report_path}')

gdbmi.exit()
```

Stuck-At Fault Injection Report

Test started: 2025-05-23 19:33:58.606208
Registers tested: pc, x1, x2, x8, x9
Iterations per mode: 2
Fault modes: stuck-at-1 then stuck-at-0 (MSB)

Iteration stuck-at-1 (MSB): 1

pc: initial=0x1000, stuck-at-1 MSB → 0x8000000000001000
x1: initial=0x0, stuck-at-1 MSB → 0x8000000000000000
x2: initial=0x0, stuck-at-1 MSB → 0x8000000000000000
x8: initial=0x0, stuck-at-1 MSB → 0x8000000000000000
x9: initial=0x0, stuck-at-1 MSB → 0x8000000000000000

Iteration stuck-at-1 (MSB): 2

pc: initial=0x8000000000001000, stuck-at-1 MSB → 0x8000000000001000
x1: initial=0x8000000000000000, stuck-at-1 MSB → 0x8000000000000000
x2: initial=0x8000000000000000, stuck-at-1 MSB → 0x8000000000000000
x8: initial=0x8000000000000000, stuck-at-1 MSB → 0x8000000000000000
x9: initial=0x-9223372036854775808, stuck-at-1 MSB → 0x-9223372036854775808

Iteration stuck-at-0 (MSB): 1

pc: initial=0x8000000000001000, stuck-at-0 MSB → 0x1000
x1: initial=0x8000000000000000, stuck-at-0 MSB → 0x0
x2: initial=0x8000000000000000, stuck-at-0 MSB → 0x0
x8: initial=0x8000000000000000, stuck-at-0 MSB → 0x0
x9: initial=0x-9223372036854775808, stuck-at-0 MSB → 0x-922b372036854775808

Iteration stuck-at-0 (MSB): 2

pc: initial=0x1000, stuck-at-0 MSB → 0x1000
x1: initial=0x0, stuck-at-0 MSB → 0x0
x2: initial=0x0, stuck-at-0 MSB → 0x0
x8: initial=0x0, stuck-at-0 MSB → 0x0
x9: initial=0x-9223372036854775808, stuck-at-0 MSB → 0x-922b372036854775808

Console Log of the Stuck-at Fault Injection Run

```
OpenSBI v1.5.1
   ____                    _____ ____ _____
  / __ \                  / ____|  _ \_   _|
 | |  | |_ __   ___ _ __ | (___ | |_) || |
 | |  | | '_ \ / _ \ '_ \ \___ \|  _ < | |
 | |__| | |_) |  __/ | | |____) | |_) || |_
  \____/| .__/ \___|_| |_|_____/|____/_____|
        | |
        |_|

Platform Name             : riscv-virtio,qemu
Platform Features         : medeleg
Platform HART Count       : 1
Platform IPI Device       : aclint-mswi
Platform Timer Device     : aclint-mtimer @ 10000000Hz
Platform Console Device   : uart8250
Platform HSM Device       : ---
Platform PMU Device       : ---
Platform Reboot Device    : syscon-reboot
Platform Shutdown Device  : syscon-poweroff
Platform Suspend Device   : ---
Platform CPPC Device      : ---
Firmware Base             : 0x80000000
Firmware Size             : 327 KB
Firmware RW Offset        : 0x40000
Firmware RW Size          : 71 KB
Firmware Heap Offset      : 0x49000
Firmware Heap Size        : 35 KB (total), 2 KB (reserved), 11 KB (used), 21 KB
(free)
Firmware Scratch Size     : 4096 B (total), 416 B (used), 3680 B (free)
Runtime SBI Version       : 2.0

Domain0 Name              : root
Domain0 Boot HART         : 0
Domain0 HARTs             : 0*
Domain0 Region00          : 0x0000000000100000-0x0000000000100fff M: (I,R,W) S/U:
(R,W)
Domain0 Region01          : 0x0000000010000000-0x0000000010000fff M: (I,R,W) S/U:
(R,W)
Domain0 Region02          : 0x0000000002000000-0x000000000200ffff M: (I,R,W) S/U:
()
Domain0 Region03          : 0x0000000080040000-0x000000008005ffff M: (R,W) S/U: ()
Domain0 Region04          : 0x0000000080000000-0x000000008003ffff M: (R,X) S/U: ()
Domain0 Region05          : 0x000000000c400000-0x000000000c5fffff M: (I,R,W) S/U:
(R,W)
Domain0 Region06          : 0x000000000c000000-0x000000000c3fffff M: (I,R,W) S/U:
(R,W)
Domain0 Region07          : 0x0000000000000000-0xffffffffffffffff M: () S/U:
(R,W,X)
Domain0 Next Address      : 0x0000000080200000
Domain0 Next Arg1         : 0x0000000087e00000
Domain0 Next Mode         : S-mode
Domain0 SysReset          : yes
Domain0 SysSuspend        : yes

Boot HART ID              : 0
Boot HART Domain          : root
Boot HART Priv Version    : v1.12
Boot HART Base ISA        : rv64imafdch
Boot HART ISA Extensions  : sstc,zicntr,zihpm,zicboz,zicbom,sdtrig,svadu
Boot HART PMP Count       : 16
Boot HART PMP Granularity : 2 bits
Boot HART PMP Address Bits: 54
```

```
Boot HART MHPM Info       : 16 (0x0007fff8)
Boot HART Debug Triggers  : 2 triggers
Boot HART MIDELEG         : 0x0000000000001666
Boot HART MEDELEG         : 0x0000000000f0b509
[    0.000000] Linux version 6.14.5 (kazumin@PR3D4T0R) (riscv64-linux-gnu-gcc
(Ubuntu 13.3.0-6ubuntu2~24.04) 13.3.0, GNU ld (GNU Binutils for Ubuntu) 2.42) #1
SMP Fri May  2 19:37:40 CEST 2025
[    0.000000] random: crng init done
[    0.000000] Machine model: riscv-virtio,qemu
[    0.000000] SBI specification v2.0 detected
[    0.000000] SBI implementation ID=0x1 Version=0x10005
[    0.000000] SBI TIME extension detected
[    0.000000] SBI IPI extension detected
[    0.000000] SBI RFENCE extension detected
[    0.000000] SBI SRST extension detected
[    0.000000] SBI DBCN extension detected
[    0.000000] efi: UEFI not found.
[    0.000000] OF: reserved mem: 0x0000000080000000..0x000000008003ffff (256 KiB)
nomap non-reusable mmode_resv1@80000000
[    0.000000] OF: reserved mem: 0x0000000080040000..0x000000008005ffff (128 KiB)
nomap non-reusable mmode_resv0@80040000
[    0.000000] Zone ranges:
[    0.000000]   DMA32    [mem 0x0000000080000000-0x0000000087ffffff]
[    0.000000]   Normal   empty
[    0.000000] Movable zone start for each node
[    0.000000] Early memory node ranges
[    0.000000]   node   0: [mem 0x0000000080000000-0x000000008005ffff]
[    0.000000]   node   0: [mem 0x0000000080060000-0x0000000087ffffff]
[    0.000000] Initmem setup node 0 [mem 0x0000000080000000-0x0000000087ffffff]
[    0.000000] SBI HSM extension detected
[    0.000000] riscv: base ISA extensions acdfhim
[    0.000000] riscv: ELF capabilities acdfim
[    0.000000] Queued spinlock using Ziccrse: enabled
[    0.000000] percpu: Embedded 22 pages/cpu s49448 r8192 d32472 u90112
[    0.000000] Kernel command line: root=/dev/vda1 console=ttyS0
[    0.000000] printk: log buffer data + meta data: 131072 + 458752 = 589824 bytes
[    0.000000] Dentry cache hash table entries: 16384 (order: 5, 131072 bytes,
linear)
[    0.000000] Inode-cache hash table entries: 8192 (order: 4, 65536 bytes, linear)
[    0.000000] Built 1 zonelists, mobility grouping on.  Total pages: 32768
[    0.000000] mem auto-init: stack:all(zero), heap alloc:off, heap free:off
[    0.000000] software IO TLB: SWIOTLB bounce buffer size adjusted to 0MB
[    0.000000] software IO TLB: area num 1.
[    0.000000] software IO TLB: mapped [mem 0x0000000087f65000-0x0000000087fa5000]
(0MB)
[    0.000000] Virtual kernel memory layout:
[    0.000000]       fixmap : 0xff1bfffffea00000 - 0xff1bffffff000000   (6144 kB)
[    0.000000]       pci io : 0xff1bffffff000000 - 0xff1c000000000000   (  16 MB)
[    0.000000]      vmemmap : 0xff1c000000000000 - 0xff20000000000000   (1024 TB)
[    0.000000]      vmalloc : 0xff20000000000000 - 0xff60000000000000   (16384 TB)
[    0.000000]      modules : 0xffffffff0158f000 - 0xffffffff80000000   (2026 MB)
[    0.000000]       lowmem : 0xff60000000000000 - 0xff60000008000000   ( 128 MB)
[    0.000000]       kernel : 0xffffffff80000000 - 0xffffffffffffffff   (2047 MB)
[    0.000000] SLUB: HWalign=64, Order=0-3, MinObjects=0, CPUs=1, Nodes=1
[    0.000000] rcu: Hierarchical RCU implementation.
[    0.000000] rcu:    RCU restricting CPUs from NR_CPUS=64 to nr_cpu_ids=1.
[    0.000000] rcu:    RCU debug extended QS entry/exit.
[    0.000000]   Tracing variant of Tasks RCU enabled.
[    0.000000] rcu: RCU calculated value of scheduler-enlistment delay is 25
jiffies.
[    0.000000] rcu: Adjusting geometry for rcu_fanout_leaf=16, nr_cpu_ids=1
[    0.000000] RCU Tasks Trace: Setting shift to 0 and lim to 1
rcu_task_cb_adjust=1 rcu_task_cpu_ids=1.
[    0.000000] NR_IRQS: 64, nr_irqs: 64, preallocated irqs: 0
[    0.000000] riscv-intc: 64 local interrupts mapped
[    0.000000] riscv: providing IPIs using SBI IPI extension
```

```
[    0.000000] rcu: srcu_init: Setting srcu_struct sizes based on contention.
[    0.000000] clocksource: riscv_clocksource: mask: 0xffffffffffffff max_cycles:
0x24e6a1710, max_idle_ns: 440795202120 ns
[    0.000072] sched_clock: 64 bits at 10MHz, resolution 100ns, wraps every
4398046511100ns
[    0.000163] riscv-timer: Timer interrupt in S-mode is available via sstc
extension
[    0.005056] Console: colour dummy device 80x25
[    0.008493] Calibrating delay loop (skipped), value calculated using timer
frequency.. 20.00 BogoMIPS (lpj=40000)
[    0.008601] pid_max: default: 32768 minimum: 301
[    0.009820] LSM: initializing lsm=capability
[    0.012076] Mount-cache hash table entries: 512 (order: 0, 4096 bytes, linear)
[    0.012107] Mountpoint-cache hash table entries: 512 (order: 0, 4096 bytes,
linear)
[    0.048904] riscv: ELF compat mode supported
[    0.049131] ASID allocator using 16 bits (65536 entries)
[    0.050295] rcu: Hierarchical SRCU implementation.
[    0.050320] rcu:    Max phase no-delay instances is 1000.
[    0.058255] EFI services will not be available.
[    0.061009] smp: Bringing up secondary CPUs ...
[    0.061484] smp: Brought up 1 node, 1 CPU
[    0.064322] Memory: 87492K/131072K available (10189K kernel code, 4941K rwdata,
4096K rodata, 2307K init, 484K bss, 41972K reserved, 0K cma-reserved)
[    0.070463] devtmpfs: initialized
[    0.080096] clocksource: jiffies: mask: 0xffffffff max_cycles: 0xffffffff,
max_idle_ns: 7645041785100000 ns
[    0.080231] futex hash table entries: 256 (order: 2, 16384 bytes, linear)
[    0.082755] pinctrl core: initialized pinctrl subsystem
[    0.088881] DMI not present or invalid.
[    0.090463] NET: Registered PF_NETLINK/PF_ROUTE protocol family
[    0.093276] DMA: preallocated 128 KiB GFP_KERNEL pool for atomic allocations
[    0.093501] DMA: preallocated 128 KiB GFP_KERNEL|GFP_DMA32 pool for atomic
allocations
[    0.093691] audit: initializing netlink subsys (disabled)
[    0.096787] thermal_sys: Registered thermal governor 'step_wise'
[    0.097456] audit: type=2000 audit(0.088:1): state=initialized audit_enabled=0
res=1
[    0.098007] cpuidle: using governor menu
[    0.122382] cpu0: Ratio of byte access time to unaligned word access is 7.06,
unaligned accesses are fast
[    0.140636] HugeTLB: registered 2.00 MiB page size, pre-allocated 0 pages
[    0.140658] HugeTLB: 28 KiB vmemmap can be freed for a 2.00 MiB page
[    0.144684] ACPI: Interpreter disabled.
[    0.145072] iommu: Default domain type: Translated
[    0.145099] iommu: DMA domain TLB invalidation policy: strict mode
[    0.146894] SCSI subsystem initialized
[    0.148311] usbcore: registered new interface driver usbfs
[    0.148519] usbcore: registered new interface driver hub
[    0.148638] usbcore: registered new device driver usb
[    0.150426] Advanced Linux Sound Architecture Driver Initialized.
[    0.160834] vgaarb: loaded
[    0.162808] clocksource: Switched to clocksource riscv_clocksource
[    0.166000] pnp: PnP ACPI: disabled
[    0.183482] NET: Registered PF_INET protocol family
[    0.184227] IP idents hash table entries: 2048 (order: 2, 16384 bytes, linear)
[    0.188851] tcp_listen_portaddr_hash hash table entries: 128 (order: 0, 4096
bytes, linear)
[    0.188931] Table-perturb hash table entries: 65536 (order: 6, 262144 bytes,
linear)
[    0.188996] TCP established hash table entries: 1024 (order: 1, 8192 bytes,
linear)
[    0.189109] TCP bind hash table entries: 1024 (order: 4, 65536 bytes, linear)
[    0.189282] TCP: Hash tables configured (established 1024 bind 1024)
[    0.189985] UDP hash table entries: 256 (order: 3, 40960 bytes, linear)
[    0.190304] UDP-Lite hash table entries: 256 (order: 3, 40960 bytes, linear)
```

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[    0.191288] NET: Registered PF_UNIX/PF_LOCAL protocol family
[    0.193460] RPC: Registered named UNIX socket transport module.
[    0.193498] RPC: Registered udp transport module.
[    0.193506] RPC: Registered tcp transport module.
[    0.193513] RPC: Registered tcp-with-tls transport module.
[    0.193520] RPC: Registered tcp NFSv4.1 backchannel transport module.
[    0.193651] PCI: CLS 0 bytes, default 64
[    0.199971] workingset: timestamp_bits=46 max_order=15 bucket_order=0
[    0.204119] NFS: Registering the id_resolver key type
[    0.204868] Key type id_resolver registered
[    0.204895] Key type id_legacy registered
[    0.205147] nfs4filelayout_init: NFSv4 File Layout Driver Registering...
[    0.205230] nfs4flexfilelayout_init: NFSv4 Flexfile Layout Driver Registering...
[    0.205711] 9p: Installing v9fs 9p2000 file system support
[    0.207181] NET: Registered PF_ALG protocol family
[    0.207446] Block layer SCSI generic (bsg) driver version 0.4 loaded (major 245)
[    0.207855] io scheduler mq-deadline registered
[    0.207922] io scheduler kyber registered
[    0.208120] io scheduler bfq registered
[    0.210861] riscv-plic: plic@c000000: mapped 95 interrupts with 1 handlers for 2
contexts.
[    0.213416] pci-host-generic 30000000.pci: host bridge /soc/pci@30000000 ranges:
[    0.213999] pci-host-generic 30000000.pci:       IO 0x0003000000..0x000300ffff -
> 0x0000000000
[    0.214446] pci-host-generic 30000000.pci:      MEM 0x0040000000..0x007fffffff -
> 0x0040000000
[    0.214509] pci-host-generic 30000000.pci:      MEM 0x0400000000..0x07ffffffff -
> 0x0400000000
[    0.214907] pci-host-generic 30000000.pci: Memory resource size exceeds max for
32 bits
[    0.215233] pci-host-generic 30000000.pci: ECAM at [mem 0x30000000-0x3fffffff]
for [bus 00-ff]
[    0.216445] pci-host-generic 30000000.pci: PCI host bridge to bus 0000:00
[    0.216694] pci_bus 0000:00: root bus resource [bus 00-ff]
[    0.216757] pci_bus 0000:00: root bus resource [io  0x0000-0xffff]
[    0.216790] pci_bus 0000:00: root bus resource [mem 0x40000000-0x7fffffff]
[    0.216799] pci_bus 0000:00: root bus resource [mem 0x400000000-0x7ffffffff]
[    0.217971] pci 0000:00:00.0: [1b36:0008] type 00 class 0x060000 conventional
PCI endpoint
[    0.221048] pci 0000:00:01.0: [1af4:1001] type 00 class 0x010000 conventional
PCI endpoint
[    0.221241] pci 0000:00:01.0: BAR 0 [io  0x0000-0x007f]
[    0.221277] pci 0000:00:01.0: BAR 1 [mem 0x00000000-0x00000fff]
[    0.221309] pci 0000:00:01.0: BAR 4 [mem 0x00000000-0x00003fff 64bit pref]
[    0.223062] pci 0000:00:01.0: BAR 4 [mem 0x400000000-0x400003fff 64bit pref]:
assigned
[    0.223248] pci 0000:00:01.0: BAR 1 [mem 0x40000000-0x40000fff]: assigned
[    0.223276] pci 0000:00:01.0: BAR 0 [io  0x0080-0x00ff]: assigned
[    0.223524] pci_bus 0000:00: resource 4 [io  0x0000-0xffff]
[    0.223554] pci_bus 0000:00: resource 5 [mem 0x40000000-0x7fffffff]
[    0.223563] pci_bus 0000:00: resource 6 [mem 0x400000000-0x7ffffffff]
[    0.227951] SBI CPPC extension NOT detected!!
[    0.234932] virtio-pci 0000:00:01.0: enabling device (0000 -> 0003)
[    0.311612] Serial: 8250/16550 driver, 4 ports, IRQ sharing disabled
[    0.321064] printk: legacy console [ttyS0] disabled
[    0.323636] 10000000.serial: ttyS0 at MMIO 0x10000000 (irq = 13, base_baud =
230400) is a 16550A
[    0.324982] printk: legacy console [ttyS0] enabled
[    1.608412] loop: module loaded
[    1.613398] virtio_blk virtio0: 1/0/0 default/read/poll queues
[    1.624514] virtio_blk virtio0: [vda] 9437184 512-byte logical blocks (4.83
GB/4.50 GiB)
[    1.653629]  vda: vda1 vda12 vda13 vda14 vda15 vda16
[    1.669033] e1000e: Intel(R) PRO/1000 Network Driver
[    1.675974] e1000e: Copyright(c) 1999 - 2015 Intel Corporation.
[    1.685746] usbcore: registered new interface driver uas
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[    1.693567] usbcore: registered new interface driver usb-storage
[    1.702544] mousedev: PS/2 mouse device common for all mice
[    1.712321] goldfish_rtc 101000.rtc: registered as rtc0
[    1.719873] goldfish_rtc 101000.rtc: setting system clock to 2025-05-24T07:34:14
UTC (1748072054)
[    1.734742] sdhci: Secure Digital Host Controller Interface driver
[    1.743399] sdhci: Copyright(c) Pierre Ossman
[    1.749050] Synopsys Designware Multimedia Card Interface Driver
[    1.757648] sdhci-pltfm: SDHCI platform and OF driver helper
[    1.766122] usbcore: registered new interface driver usbhid
[    1.773964] usbhid: USB HID core driver
[    1.779735] riscv-pmu-sbi: SBI PMU extension is available
[    1.787352] riscv-pmu-sbi: 16 firmware and 18 hardware counters
[    1.795668] riscv-pmu-sbi: Perf sampling/filtering is not supported as sscof
extension is not available
[    1.813383] NET: Registered PF_INET6 protocol family
[    1.825570] Segment Routing with IPv6
[    1.830820] In-situ OAM (IOAM) with IPv6
[    1.836173] sit: IPv6, IPv4 and MPLS over IPv4 tunneling driver
[    1.846352] NET: Registered PF_PACKET protocol family
[    1.854115] 9pnet: Installing 9P2000 support
[    1.860291] Key type dns_resolver registered
[    1.897803] debug_vm_pgtable: [debug_vm_pgtable         ]: Validating
architecture page table helpers
[    1.918725] clk: Disabling unused clocks
[    1.924133] PM: genpd: Disabling unused power domains
[    1.930973] ALSA device list:
[    1.934845]   No soundcards found.
[    1.964119] EXT4-fs (vda1): INFO: recovery required on readonly filesystem
[    1.973578] EXT4-fs (vda1): write access will be enabled during recovery
[    2.000348] EXT4-fs (vda1): recovery complete
[    2.009101] EXT4-fs (vda1): mounted filesystem 626ef4b6-84d2-4b70-bd20-
e9f892339465 ro with ordered data mode. Quota mode: disabled.
[    2.027321] VFS: Mounted root (ext4 filesystem) readonly on device 254:1.
[    2.039681] devtmpfs: mounted
[    2.083204] Freeing unused kernel image (initmem) memory: 2304K
[    2.089829] Run /sbin/init as init process
[    2.359833] systemd[1]: systemd 255.4-1ubuntu8.5 running in system mode (+PAM
+AUDIT +SELINUX +APPARMOR +IMA +SMACK +SECCOMP +GCRYPT -GNUTLS +OPENSSL +ACL +BLKID
+CURL +ELFUTILS +FIDO2 +IDN2 -IDN +IPTC +KMOD +LIBCRYPTSETUP +LIBFDISK +PCRE2 -
PWQUALITY +P11KIT +QRENCODE +TPM2 +BZIP2 +LZ4 +XZ +ZLIB +ZSTD -BPF_FRAMEWORK -
XKBCOMMON +UTMP +SYSVINIT default-hierarchy=unified)
[    2.396461] systemd[1]: Detected virtualization qemu.
[    2.402372] systemd[1]: Detected architecture riscv64.

Welcome to [1mUbuntu 24.04.2 LTS[0m!


[    2.423764] systemd[1]: Hostname set to <ubuntu>.
[    4.527622] systemd[1]: Queued start job for default target graphical.target.
[    4.589167] systemd[1]: Created slice system-modprobe.slice - Slice
/system/modprobe.
[[0;32m  OK  [0m] Created slice [0;1;39msystem-modprobe.slice[0m - Slice
/system/modprobe.

[    4.617685] systemd[1]: Created slice system-serial\x2dgetty.slice - Slice
/system/serial-getty.
[[0;32m  OK  [0m] Created slice [0;1;39msystem-serial\x2dgetâ€¦slice[0m - Slice
/system/serial-getty.

[    4.647060] systemd[1]: Created slice system-systemd\x2dfsck.slice - Slice
/system/systemd-fsck.
[[0;32m  OK  [0m] Created slice [0;1;39msystem-systemd\x2dfsâ€¦slice[0m - Slice
/system/systemd-fsck.

[    4.674485] systemd[1]: Created slice user.slice - User and Session Slice.
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[[0;32m  OK  [0m] Created slice [0;1;39muser.slice[0m - User and Session Slice.

[    4.695910] systemd[1]: Started systemd-ask-password-wall.path - Forward
Password Requests to Wall Directory Watch.
[[0;32m  OK  [0m] Started [0;1;39msystemd-ask-password-wall.â€¦[0md Requests to
Wall Directory Watch.

[    4.724237] systemd[1]: proc-sys-fs-binfmt_misc.automount - Arbitrary Executable
File Formats File System Automount Point was skipped because of an unmet condition
check (ConditionPathExists=/proc/sys/fs/binfmt_misc).
[    4.749751] systemd[1]: Expecting device dev-disk-by\x2dlabel-UEFI.device -
/dev/disk/by-label/UEFI...
         Expecting device [0;1;39mdev-disk-by\x2dlaâ€¦device[0m - /dev/disk/by-
label/UEFI...

[    4.774307] systemd[1]: Expecting device dev-ttyS0.device - /dev/ttyS0...
         Expecting device [0;1;39mdev-ttyS0.device[0m - /dev/ttyS0...

[    4.792474] systemd[1]: Reached target integritysetup.target - Local Integrity
Protected Volumes.
[[0;32m  OK  [0m] Reached target [0;1;39mintegritysetup.targâ€¦[0m Local Integrity
Protected Volumes.

[    4.816935] systemd[1]: Reached target slices.target - Slice Units.
[[0;32m  OK  [0m] Reached target [0;1;39mslices.target[0m - Slice Units.

[    4.835417] systemd[1]: Reached target snapd.mounts-pre.target - Mounting snaps.
[[0;32m  OK  [0m] Reached target [0;1;39msnapd.mounts-pre.target[0m - Mounting
snaps.

[    4.856492] systemd[1]: Reached target snapd.mounts.target - Mounted snaps.
[[0;32m  OK  [0m] Reached target [0;1;39msnapd.mounts.target[0m - Mounted snaps.

[    4.876505] systemd[1]: Reached target swap.target - Swaps.
[[0;32m  OK  [0m] Reached target [0;1;39mswap.target[0m - Swaps.

[    4.892837] systemd[1]: Reached target veritysetup.target - Local Verity
Protected Volumes.
[[0;32m  OK  [0m] Reached target [0;1;39mveritysetup.target[0m - Local Verity
Protected Volumes.

[    4.918604] systemd[1]: Listening on dm-event.socket - Device-mapper event
daemon FIFOs.
[[0;32m  OK  [0m] Listening on [0;1;39mdm-event.socket[0m - Device-mapper event
daemon FIFOs.

[    4.942782] systemd[1]: Listening on lvm2-lvmpolld.socket - LVM2 poll daemon
socket.
[[0;32m  OK  [0m] Listening on [0;1;39mlvm2-lvmpolld.socket[0m - LVM2 poll daemon
socket.

[    4.965748] systemd[1]: Listening on multipathd.socket - multipathd control
socket.
[[0;32m  OK  [0m] Listening on [0;1;39mmultipathd.socket[0m - multipathd control
socket.

[    4.988256] systemd[1]: Listening on syslog.socket - Syslog Socket.
[[0;32m  OK  [0m] Listening on [0;1;39msyslog.socket[0m - Syslog Socket.

[    5.006717] systemd[1]: Listening on systemd-fsckd.socket - fsck to fsckd
communication Socket.
[[0;32m  OK  [0m] Listening on [0;1;39msystemd-fsckd.socket[0m â€¦fsck to fsckd
communication Socket.

[    5.031717] systemd[1]: Listening on systemd-initctl.socket - initctl
Compatibility Named Pipe.
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[[0;32m  OK  [0m] Listening on [0;1;39msystemd-initctl.sockeâ€¦[0m- initctl
Compatibility Named Pipe.

[    5.057001] systemd[1]: Listening on systemd-journald-dev-log.socket - Journal
Socket (/dev/log).
[[0;32m  OK  [0m] Listening on [0;1;39msystemd-journald-dev-â€¦socket[0m - Journal
Socket (/dev/log).

[    5.083674] systemd[1]: Listening on systemd-journald.socket - Journal Socket.
[[0;32m  OK  [0m] Listening on [0;1;39msystemd-journald.socket[0m - Journal Socket.

[    5.105864] systemd[1]: Listening on systemd-networkd.socket - Network Service
Netlink Socket.
[[0;32m  OK  [0m] Listening on [0;1;39msystemd-networkd.socket[0m - Network Service
Netlink Socket.

[    5.129965] systemd[1]: systemd-pcrextend.socket - TPM2 PCR Extension (Varlink)
was skipped because of an unmet condition check (ConditionSecurity=measured-uki).
[    5.150778] systemd[1]: Listening on systemd-udevd-control.socket - udev Control
Socket.
[[0;32m  OK  [0m] Listening on [0;1;39msystemd-udevd-control.socket[0m - udev
Control Socket.

[    5.174089] systemd[1]: Listening on systemd-udevd-kernel.socket - udev Kernel
Socket.
[[0;32m  OK  [0m] Listening on [0;1;39msystemd-udevd-kernel.socket[0m - udev Kernel
Socket.

[    5.223908] systemd[1]: Mounting dev-hugepages.mount - Huge Pages File System...
         Mounting [0;1;39mdev-hugepages.mount[0m - Huge Pages File System...

[    5.287000] systemd[1]: Mounting dev-mqueue.mount - POSIX Message Queue File
System...
         Mounting [0;1;39mdev-mqueue.mount[0m - POSIX Message Queue File System...

[    5.382980] systemd[1]: Mounting sys-kernel-debug.mount - Kernel Debug File
System...
         Mounting [0;1;39msys-kernel-debug.mount[0m - Kernel Debug File System...

[    5.424962] systemd[1]: sys-kernel-tracing.mount - Kernel Trace File System was
skipped because of an unmet condition check
(ConditionPathExists=/sys/kernel/tracing).
[    5.496825] systemd[1]: Starting systemd-journald.service - Journal Service...
         Starting [0;1;39msystemd-journald.service[0m - Journal Service...

[    5.571216] systemd[1]: Starting keyboard-setup.service - Set the console
keyboard layout...
         Starting [0;1;39mkeyboard-setup.service[0m - Set the console keyboard
layout...

[    5.621267] systemd[1]: kmod-static-nodes.service - Create List of Static Device
Nodes was skipped because of an unmet condition check
(ConditionFileNotEmpty=/lib/modules/6.14.5/modules.devname).
[    5.699359] systemd[1]: Starting lvm2-monitor.service - Monitoring of LVM2
mirrors, snapshots etc. using dmeventd or progress polling...
         Starting [0;1;39mlvm2-monitor.service[0m - Moâ€¦ing dmeventd or progress
polling...

[    5.872109] systemd[1]: Starting modprobe@configfs.service - Load Kernel Module
configfs...
         Starting [0;1;39mmodprobe@configfs.service[0m - Load Kernel Module
configfs...

[    5.985581] systemd-journald[88]: Collecting audit messages is disabled.
[    6.063406] systemd[1]: Starting modprobe@dm_mod.service - Load Kernel Module
dm_mod...
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         Starting [0;1;39mmodprobe@dm_mod.service[0m - Load Kernel Module dm_mod...

[    6.247349] systemd[1]: Starting modprobe@drm.service - Load Kernel Module
drm...
         Starting [0;1;39mmodprobe@drm.service[0m - Load Kernel Module drm...

[    6.407620] systemd[1]: Starting modprobe@efi_pstore.service - Load Kernel
Module efi_pstore...
         Starting [0;1;39mmodprobe@efi_pstore.serviâ€¦[0m - Load Kernel Module
efi_pstore...

[    6.595398] systemd[1]: Starting modprobe@fuse.service - Load Kernel Module
fuse...
         Starting [0;1;39mmodprobe@fuse.service[0m - Load Kernel Module fuse...

[    6.723366] systemd[1]: Starting modprobe@loop.service - Load Kernel Module
loop...
         Starting [0;1;39mmodprobe@loop.service[0m - Load Kernel Module loop...

[    6.789457] systemd[1]: netplan-ovs-cleanup.service - OpenVSwitch configuration
for cleanup was skipped because of an unmet condition check
(ConditionFileIsExecutable=/usr/bin/ovs-vsctl).
[    6.887494] systemd[1]: Starting systemd-fsck-root.service - File System Check
on Root Device...
         Starting [0;1;39msystemd-fsck-root.service[0mâ€¦File System Check on Root
Device...

[    7.003401] systemd[1]: Starting systemd-modules-load.service - Load Kernel
Modules...
         Starting [0;1;39msystemd-modules-load.service[0m - Load Kernel Modules...

[    7.098861] systemd[1]: systemd-pcrmachine.service - TPM2 PCR Machine ID
Measurement was skipped because of an unmet condition check
(ConditionSecurity=measured-uki).
[    7.231350] systemd[1]: Starting systemd-tmpfiles-setup-dev-early.service -
Create Static Device Nodes in /dev gracefully...
         Starting [0;1;39msystemd-tmpfiles-setup-deâ€¦[0m Device Nodes in /dev
gracefully...

[    7.346947] systemd[1]: systemd-tpm2-setup-early.service - TPM2 SRK Setup
(Early) was skipped because of an unmet condition check
(ConditionSecurity=measured-uki).
[    7.439420] systemd[1]: Starting systemd-udev-trigger.service - Coldplug All
udev Devices...
         Starting [0;1;39msystemd-udev-trigger.service[0m - Coldplug All udev
Devices...

[    7.593293] systemd[1]: Started systemd-journald.service - Journal Service.
[[0;32m  OK  [0m] Started [0;1;39msystemd-journald.service[0m - Journal Service.

[[0;32m  OK  [0m] Mounted [0;1;39mdev-hugepages.mount[0m - Huge Pages File System.

[[0;32m  OK  [0m] Mounted [0;1;39mdev-mqueue.mount[0m - POSIX Message Queue File
System.

[[0;32m  OK  [0m] Mounted [0;1;39msys-kernel-debug.mount[0m - Kernel Debug File
System.

[[0;32m  OK  [0m] Finished [0;1;39mkeyboard-setup.service[0m - Set the console
keyboard layout.

[[0;32m  OK  [0m] Finished [0;1;39mlvm2-monitor.service[0m - Moâ€¦using dmeventd or
progress polling.
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[[0;32m  OK  [0m] Finished [0;1;39mmodprobe@configfs.service[0m - Load Kernel
Module configfs.

[[0;32m  OK  [0m] Finished [0;1;39mmodprobe@dm_mod.service[0m - Load Kernel Module
dm_mod.

[[0;32m  OK  [0m] Finished [0;1;39mmodprobe@drm.service[0m - Load Kernel Module
drm.

[[0;32m  OK  [0m] Finished [0;1;39mmodprobe@efi_pstore.service[0m - Load Kernel
Module efi_pstore.

[[0;32m  OK  [0m] Finished [0;1;39mmodprobe@fuse.service[0m - Load Kernel Module
fuse.

[[0;32m  OK  [0m] Finished [0;1;39mmodprobe@loop.service[0m - Load Kernel Module
loop.

[[0;32m  OK  [0m] Finished [0;1;39msystemd-fsck-root.service[0m - File System Check
on Root Device.

[[0;32m  OK  [0m] Finished [0;1;39msystemd-modules-load.service[0m - Load Kernel
Modules.

[[0;32m  OK  [0m] Finished [0;1;39msystemd-tmpfiles-setup-deâ€¦[0mic Device Nodes
in /dev gracefully.

[[0;32m  OK  [0m] Started [0;1;39msystemd-fsckd.service[0m - Fiâ€¦stem Check Daemon
to report status.

         Starting [0;1;39msystemd-remount-fs.servicâ€¦[0munt Root and Kernel File
Systems...

         Starting [0;1;39msystemd-sysctl.service[0m - Apply Kernel Variables...

[[0;32m  OK  [0m] Finished [0;1;39msystemd-sysctl.service[0m - Apply Kernel
Variables.

[[0;32m  OK  [0m] Finished [0;1;39msystemd-remount-fs.servicâ€¦[0mmount Root and
Kernel File Systems.

         Starting [0;1;39mcloud-init-local.service[0m â€¦-init: Local Stage (pre-
network)...

         Starting [0;1;39mmultipathd.service[0m - Deviâ€¦pper Multipath Device
Controller...

         Starting [0;1;39msystemd-journal-flush.serâ€¦[0msh Journal to Persistent
Storage...

         Starting [0;1;39msystemd-random-seed.service[0m - Load/Save OS Random
Seed...

         Starting [0;1;39msystemd-tmpfiles-setup-deâ€¦[0meate Static Device Nodes
in /dev...

[[0;32m  OK  [0m] Finished [0;1;39msystemd-random-seed.service[0m - Load/Save OS
Random Seed.

[[0;32m  OK  [0m] Finished [0;1;39msystemd-tmpfiles-setup-deâ€¦[0mCreate Static
Device Nodes in /dev.

         Starting [0;1;39msystemd-udevd.service[0m - Râ€¦ager for Device Events and
Files...
```

[[0;32m OK [0m] Finished [0;1;39msystemd-journal-flush.serâ€¦[0mlush Journal to Persistent Storage.

[[0;1;31mFAILED[0m] Failed to start [0;1;39mmultipathd.service[0mâ€¦Mapper Multipath Device Controller.

See 'systemctl status multipathd.service' for details.

[[0;32m OK [0m] Reached target [0;1;39mlocal-fs-pre.target[0mâ€¦Preparation for Local File Systems.

[[0;32m OK [0m] Started [0;1;39msystemd-udevd.service[0m - Ruâ€¦anager for Device Events and Files.

[[0;32m OK [0m] Finished [0;1;39msystemd-udev-trigger.service[0m - Coldplug All udev Devices.

[[0;32m OK [0m] Found device [0;1;39mdev-ttyS0.device[0m - /dev/ttyS0.

[[0;32m OK [0m] Found device [0;1;39mdev-disk-by\x2dlabel-â€¦I.device[0m - /dev/disk/by-label/UEFI.

[[0;32m OK [0m] Started [0;1;39msystemd-ask-password-consoâ€¦[0mequests to Console Directory Watch.

[[0;32m OK [0m] Reached target [0;1;39mcryptsetup.target[0m - Local Encrypted Volumes.

Starting [0;1;39msystemd-fsck@dev-disk-by\â€¦[0mCheck on /dev/disk/by-label/UEFI...

[[0;32m OK [0m] Finished [0;1;39msystemd-fsck@dev-disk-by\â€¦[0mm Check on /dev/disk/by-label/UEFI.

Mounting [0;1;39mboot-efi.mount[0m - /boot/efi...

[24.546480] FAT-fs (vda15): IO charset iso8859-1 not found
[[0;1;31mFAILED[0m] Failed to mount [0;1;39mboot-efi.mount[0m - /boot/efi.

See 'systemctl status boot-efi.mount' for details.

[[0;1;38;5;185mDEPEND[0m] Dependency failed for [0;1;39mlocal-fs.target[0m - Local File Systems.

[[0;32m OK [0m] Stopped [0;1;39msystemd-ask-password-consoâ€¦[0mequests to Console Directory Watch.

[[0;32m OK [0m] Stopped [0;1;39msystemd-ask-password-wall.â€¦[0md Requests to Wall Directory Watch.

[[0;32m OK [0m] Reached target [0;1;39mtimers.target[0m - Timer Units.

[[0;32m OK [0m] Listening on [0;1;39msystemd-sysext.socket[0mâ€¦tension Image Management (Varlink).

[[0;32m OK [0m] Reached target [0;1;39mcloud-init.target[0m - Cloud-init target.

Starting [0;1;39mconsole-setup.service[0m - Set console font and keymap...

Starting [0;1;39mfinalrd.service[0m - Create â€¦time dir for shutdown pivot root...

[[0;32m OK [0m] Closed [0;1;39msyslog.socket[0m - Syslog Socket.

[[0;32m OK [0m] Reached target [0;1;39mgetty.target[0m - Login Prompts.

[[0;32m OK [0m] Reached target [0;1;39mgetty-pre.target[0m - Preparation for Logins.

         Starting [0;1;39msetvtrgb.service[0m - Set console scheme...

         Starting [0;1;39mufw.service[0m - Uncomplicated firewall...

[[0;32m OK [0m] Reached target [0;1;39mpaths.target[0m - Path Units.

[[0;32m OK [0m] Reached target [0;1;39msockets.target[0m - Socket Units.

[[0;32m OK [0m] Listening on [0;1;39mdbus.socket[0m - D-Bus System Message Bus Socket.

         Starting [0;1;39mdbus.service[0m - D-Bus System Message Bus...

[[0;32m OK [0m] Started [0;1;39memergency.service[0m - Emergency Shell.

[[0;32m OK [0m] Reached target [0;1;39memergency.target[0m - Emergency Mode.

         Starting [0;1;39mplymouth-read-write.serviâ€¦[0mymouth To Write Out Runtime Data...

         Starting [0;1;39msystemd-tmpfiles-setup.seâ€¦[0me Volatile Files and Directories...

[[0;32m OK [0m] Finished [0;1;39mconsole-setup.service[0m - Set console font and keymap.

[[0;32m OK [0m] Finished [0;1;39mfinalrd.service[0m - Create â€¦untime dir for shutdown pivot root.

[[0;32m OK [0m] Finished [0;1;39msetvtrgb.service[0m - Set console scheme.

[[0;32m OK [0m] Finished [0;1;39mufw.service[0m - Uncomplicated firewall.

[[0;32m OK [0m] Finished [0;1;39mplymouth-read-write.serviâ€¦[0mPlymouth To Write Out Runtime Data.

[[0;32m OK [0m] Started [0;1;39mdbus.service[0m - D-Bus System Message Bus.

[[0;32m OK [0m] Created slice [0;1;39msystem-getty.slice[0m - Slice /system/getty.

[[0;32m OK [0m] Finished [0;1;39msystemd-tmpfiles-setup.seâ€¦[0mate Volatile Files and Directories.

         Starting [0;1;39msystemd-resolved.service[0m - Network Name Resolution...

         Starting [0;1;39msystemd-timesyncd.service[0m - Network Time Synchronization...

         Starting [0;1;39msystemd-update-utmp.serviâ€¦[0mord System Boot/Shutdown in UTMP...

[[0;32m OK [0m] Finished [0;1;39msystemd-update-utmp.serviâ€¦[0mecord System Boot/Shutdown in UTMP.

         Starting [0;1;39msystemd-update-utmp-runleâ€¦[0m- Record Runlevel Change in UTMP...

[[0;32m OK [0m] Finished [0;1;39msystemd-update-utmp-runleâ€¦e[0m - Record Runlevel Change in UTMP.

```
[[0;32m  OK  [0m] Started [0;1;39msystemd-timesyncd.service[0m - Network Time Synchronization.

[[0;32m  OK  [0m] Reached target [0;1;39mtime-set.target[0m - System Time Set.

[[0;32m  OK  [0m] Started [0;1;39msystemd-resolved.service[0m - Network Name Resolution.

[[0;32m  OK  [0m] Reached target [0;1;39mnss-lookup.target[0m - Host and Network Name Lookups.

[   30.267317] cloud-init[247]: Cloud-init v. 24.4-0ubuntu1~24.04.2 running 'init-local' at Sat, 24 May 2025 07:34:42 +0000. Up 29.54 seconds.
You are in emergency mode. After logging in, type "journalctl -xb" to view
system logs, "systemctl reboot" to reboot, or "exit"
to continue bootup.
Press Enter for maintenance
(or press Control-D to continue):
```

Console Log of the Instruction Corruption Fault Injection Run

```
from pygdbmi.gdbcontroller import GdbController
from docx import Document
from datetime import datetime
import time

# Configurable parameters
BASE_ADDR = 0xFFFFFFFF80000000  # Adjust to actual kernel code base
RANGE = 64  # Number of instructions to flip
FLIP_MASK = 1 << 31  # MSB flip
REPORT_PATH = "/mnt/f/loggins/instruction_flip_msb_report.docx"

# Initialize
doc = Document()
doc.add_heading("Instruction Bit Flip Injection Report", 0)
doc.add_paragraph(f"Started: {datetime.now()}")
doc.add_paragraph(f"Base Address: {hex(BASE_ADDR)}")
doc.add_paragraph(f"Bit flipped: MSB (bit 31)")
doc.add_paragraph(f"Range: {RANGE} instructions")
doc.add_paragraph("")

gdbmi = GdbController(command=["gdb-multiarch", "--nx", "--interpreter=mi3"])

# Retry connection logic
connected = False
for attempt in range(10):
   try:
      gdbmi.write("-target-select remote localhost:1234")
      time.sleep(1)
      response = gdbmi.get_gdb_response(timeout_sec=2, raise_error_on_timeout=False)
      if any("connected" in r.get("payload", "") for r in response):
         connected = True
         break
   except:
      pass
   print(f"[!] GDB not ready (attempt {attempt+1}), retrying...")
   time.sleep(1)

if not connected:
   print("[-] Could not connect to QEMU GDB stub. Exiting.")
   gdbmi.exit()
   exit(1)

# Injection loop
for i in range(RANGE):
   addr = BASE_ADDR + i * 4
   gdbmi.write(f"x /1wx {hex(addr)}")
   response = gdbmi.get_gdb_response(timeout_sec=2, raise_error_on_timeout=False)

   original = None
   for r in response:
      if r["type"] == "console" and r["payload"].startswith("0x"):
         try:
            original = int(r["payload"].split(":")[1], 16)
```

```
                break
            except:
                continue

    if original is None:
        doc.add_paragraph(f"[{i}] Failed to read at {hex(addr)}")
        continue

    corrupted = original ^ FLIP_MASK
    gdbmi.write(f"set {hex(addr)} = {hex(corrupted)}")
    doc.add_paragraph(f"[{i}] {hex(addr)}: {hex(original)} -> {hex(corrupted)}")

# Save and exit
doc.save(REPORT_PATH)
gdbmi.exit()
print("[+] Instruction corruption complete. Report saved.")
```